TB Ring Modulation

DETAILED ENGLISH USER MANUAL

A streamlined internal-carrier ring modulator with Frequency, Shape, Tone, Mix, smoothing, and factory programs.

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Host-visible endpoints documented: 6

1. Product purpose

A streamlined internal-carrier ring modulator with Frequency, Shape, Tone, Mix, smoothing, and factory programs.

Ring modulation is powerful but often slowed by external carrier routing. TB Ring Modulation supplies its own carrier from 0.1 Hz to 5 kHz, covering tremolo, pulses, robotic speech, bell-like sidebands, and extreme metallic textures with four primary sound controls.

What result it creates

- Slow tremolo and pulsing amplitude movement
- Metallic sum-and-difference sidebands
- Robotic voices and tuned percussion
- Simple automation without external routing

Signal flow

Input x internal carrier oscillator -> carrier Shape -> Tone filtering -> Mix -> output

2. Complete feature guide

Frequency

Below the audible range, Frequency behaves like tremolo. As it enters the audio range, new sum-and-difference sidebands become the main effect.

Shape

Moves the carrier waveform from Sine toward progressively harder shapes in the deployed control mapping. Smoother shapes produce cleaner sidebands; harder shapes add more harmonic families.

Tone

Sets the upper spectral focus from 500 Hz to 20 kHz, controlling edge and brightness after modulation.

Mix

Blends the ring-modulated signal with the original. Partial Mix preserves pitch identity; full wet exposes the sideband structure.

Programs and smoothing

Programs cover Subtle Tremolo, Bell Alloy, Robot Voice, Radio Sidebands, and Digital Insect. Parameter and bypass smoothing reduce clicks during automation.

3. Quick start

1. Set Mix to 100 percent to learn the effect.

2. Sweep Frequency from sub-audio into the audible range.

3. Choose Shape for sideband density.

4. Use Tone to control harshness.

5. Reduce Mix to restore the original note.

6. Automate Frequency slowly for transitions or rhythmically for designed motion.

4. Practical workflows

Subtle tremolo

1. Set Frequency below about 20 Hz.

2. Use a smooth Shape.

3. Set Mix to taste and leave Tone open.

Expected result: The level pulses without generating a strongly metallic spectrum.

Robot voice

1. Set Frequency in the low-to-mid audio range.

2. Use a harder Shape.

3. Lower Tone if consonants become sharp.

4. Keep Mix high.

Expected result: Speech retains rhythm while pitch becomes mechanical and sideband-rich.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Ring modulation is inharmonic unless carrier and source relationships align.
- High frequencies and hard shapes can be bright.
- The internal carrier does not track source pitch.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 6 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 773352680 Source ID bypass	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Frequency VST3 ID 2077459804 Source ID frequency	0.10 to 5000	120	Automatable	Sets the principal frequency, note, or spectral center used by this function.
Mix VST3 ID 108124 Source ID mix	0.0 to 100.0	50.0	Automatable	Sets the balance between the original signal and the complete processed path.
Program VST3 ID 1886548852 Source ID program	Init, Subtle Tremolo, Bell Alloy, Robot Voice, Radio Sidebands, Digital Insect	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Shape VST3 ID 109399969 Source ID shape	Sine to Square	Sine	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Tone VST3 ID 3565938 Source ID tone	500 to 20000	12000	Automatable	Host-automatable control for Tone. Its full range and default are shown in this table; use it with the related feature section above.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.